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# OBSERVATION WELL RESPONSE TIME AND ITS EFFECT UPON AQUIFER TEST RESULTS

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ABSTRACT

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The concept of observation-well response time is introduced in the context of aquifer pump tests. Previous work on hydrostatic time lag and slug tests is considered together with the precise way in which an observation well responds during an aquifer test. Using the concepts of Hvorslev an equation is produced linking observation-well response time, $a$, and a dimensionless response-time factor $\beta$. The Theis equation is then modified by various values of $\beta$ to derive a set of type curves. The two parameters $a$ and $\beta$ are related by a simple equation involving aquifer transmissivity, $T$, and storativity, $S$. Both $\alpha$ and $\beta$ can be obtained from an aquifer test whilst $a$ alone can be obtained independently from a slug test. An example is given showing the effect of $\beta$ upon aquifer test results and the value of $a$ calculated from the aquifer test analysis is corroborated by slug test results. The definition of $\beta$ is used to show in what circumstances it will be important and how its effect can be reduced to a minimum. The introduction of slug testing of observation wells as standard practice for all aquifer tests is suggested.

## INTRODUCTION

The analysis of drawdowns resulting from groundwater abstraction to yield estimates of aquifer properties — aquifer test analysis — has become a basic tool of hydrogeology. Analytical techniques have been developed to take account of a wide range of aquifer conditions and modes of pumping and yet the influence of possible response delays of water levels within observation wells upon results has hitherto been largely ignored although it was considered briefly by Stallman (1965) using an electrical analogue model. It has been assumed that observation-well water levels respond instantly with changing aquifer piezometric levels whereas these levels must respond as a result of changing aquifer pressures with, as a consequence, delays in response. This disregard for the influence of observation-well response delays upon the results of aquifer test analysis has been due, inter alia, to the lack of methods for the accurate measurement of well as opposed to aquifer constants and the opinion that observation-well response-delay effects are of secondary importance compared to other test variables.

Since the work of Hvorslev (1951) on "hydrostatic time lag" and Bredehoeft et al. (1966), Cooper et al. (1967) and Papadopulos et al. (1973), on the response of a finite-diameter well to an instantaneous charge of water it has been possible to measure constants which apply strictly to a well, its storage, its "hydrostatic time lag" and its local hydrogeological environment. Thus it is now possible to ascribe to a particular well local values of transmissivity, $T$, and storativity, $S$, which will probably differ from the values calculated for the aquifer. However, even if the localized values of $T$ and $S$ are the same as those of the aquifer this does not detract from the fact that every observation well needs a finite time ("hydrostatic time lag" or well response time) to respond to a change in aquifer piezometric level. It is the purpose of this paper to show how response delays in observation wells can affect aquifer test results and to present a method to determine the magnitude of such effects.

## DERIVATION OF TYPE CURVES WITH OBSERVATION-WELL RESPONSE DELAY

Before the start of a pumping test on a non-leaky confined aquifer it is assumed that the well water level is coincident with the aquifer piezometric level (Fig.1A). Aquifer drawdowns will be described by the Theis equation:

$$s = (Q/4\pi T)\, W(u) \tag{1}$$

where

$$u = r_0^2 S/4Tt \tag{1a}$$

and

$$W(u) = \int_u^\infty (1/\lambda) \exp(-\lambda) d\lambda \tag{1b}$$

whilst the rate of drawdown will be given by the first derivative, i.e.:

$$ds/dt = (-Q/\pi r_0^2 S)\, u \exp(-u) \tag{2}$$

Eq. 2 shows that at the start of a pumping test the rate of drawdown will be small at first, later increasing to a maximum when, according to the Theis analysis, $u = 1$. For example, using eq. 1a, for a pumping test where $S = 10^{-3}$, $T = 200\ \mathrm{m^2\ day^{-1}}$ and $r_0 = 100$ m the time of maximum rate of drawdown corresponds to 18 min after the start of pumping. Water will flow out of the observation well in response to the pressure difference between the well and aquifer levels (Fig.1B). The greater the rate of drawdown, the greater will be the difference between aquifer (i.e. real) levels and well or measured water levels stemming from the response time lag of the observation well. As the rate of drawdown slows, the observation-well levels will approach the aquifer levels again and effectively coincide in the limit of long pumping times (Fig.1C). This description may seem obvious but the universal use of log-log and semi-log curves tends to disguise the rates of change of water levels.

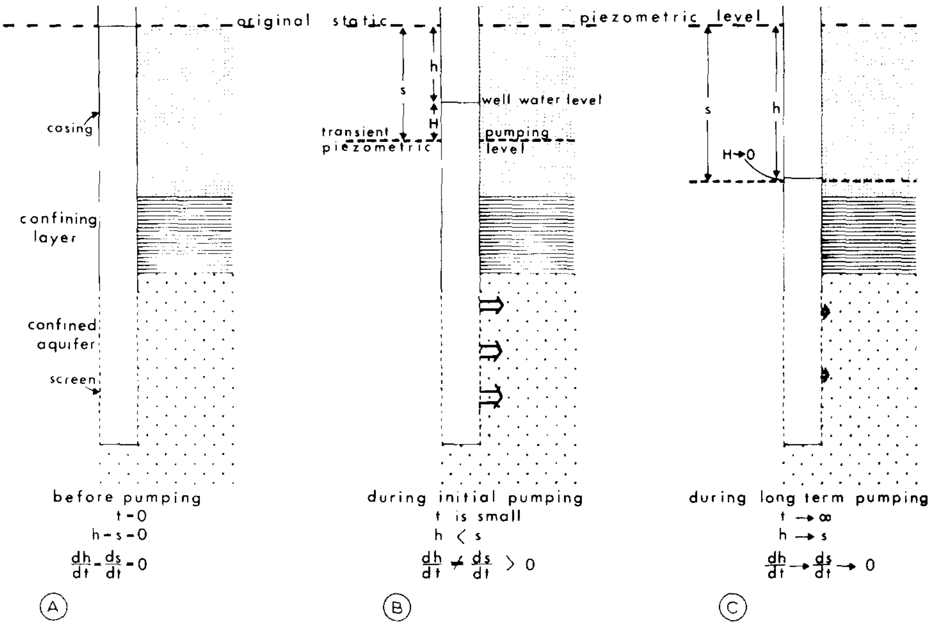


Fig.1. Variation of aquifer and well water levels during a pumping test.
A. Before pumping; B. during initial pumping; and C. after long-term pumping.

If the observation well is now considered to have a response time lag, a modified Theis equation can be derived which links the observation-well measured drawdown, $h$, with the actual aquifer drawdown, $s$:

$$s = h + H \tag{3}$$

Using the concepts of Hvorslev (1951) to relate the flow rate between the observation well and the aquifer as proportional to the difference in head, $H$, and neglecting inertial effects then:

$$Q' = FKH \tag{4}$$

The equation of flow for the observation well is:

$$Q' = \pi r_w^2 (\mathrm{d}h/\mathrm{d}t) \tag{5}$$

Combining eqs. 3—5 gives:

$$\mathrm{d}h/\mathrm{d}t + FKh/\pi r_W^2 = (FK/\pi r_W^2)\, s(t) \tag{6}$$

with initial condition: at $t = 0$, $h = 0$ and $s = 0$.

The solution to this first-order ordinary differential equation can be written as:

$$h(t) = (1/a) \exp(-t/a) \int_0^t \exp(\eta/a) s(\eta)\, \mathrm{d}\eta \tag{7}$$

TABLE I

Values of $W^*(u, \beta)$ for various $\beta$

| | | $W^*(u,\beta)$ for various $\beta$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $1/u$ | $W(u)$ | 0.1 | 0.5 | 1.0 | 2.0 | 5.0 | 10 | 20 | 50 | 100 |
| 0.1 | $4.16 \cdot 10^{-6}$ | $3.05 \cdot 10^{-7}$ | $6.45 \cdot 10^{-8}$ | $3.24 \cdot 10^{-8}$ | $1.63 \cdot 10^{-8}$ | $6.53 \cdot 10^{-9}$ | $3.27 \cdot 10^{-9}$ | $1.63 \cdot 10^{-9}$ | $6.53 \cdot 10^{-10}$ | $3.27 \cdot 10^{-10}$ |
| 0.15 | $1.68 \cdot 10^{-4}$ | $2.38 \cdot 10^{-5}$ | $5.28 \cdot 10^{-6}$ | $2.68 \cdot 10^{-6}$ | $1.35 \cdot 10^{-6}$ | $5.42 \cdot 10^{-7}$ | $2.71 \cdot 10^{-7}$ | $1.36 \cdot 10^{-7}$ | $5.43 \cdot 10^{-8}$ | $2.71 \cdot 10^{-8}$ |
| 0.2 | $1.15 \cdot 10^{-3}$ | $2.48 \cdot 10^{-4}$ | $5.82 \cdot 10^{-5}$ | $2.97 \cdot 10^{-5}$ | $1.50 \cdot 10^{-5}$ | $6.05 \cdot 10^{-6}$ | $3.03 \cdot 10^{-6}$ | $1.52 \cdot 10^{-6}$ | $6.07 \cdot 10^{-7}$ | $3.04 \cdot 10^{-7}$ |
| 0.25 | $3.78 \cdot 10^{-3}$ | $1.10 \cdot 10^{-3}$ | $2.73 \cdot 10^{-4}$ | $1.41 \cdot 10^{-4}$ | $7.15 \cdot 10^{-5}$ | $2.89 \cdot 10^{-5}$ | $1.45 \cdot 10^{-5}$ | $7.25 \cdot 10^{-6}$ | $2.90 \cdot 10^{-6}$ | $1.45 \cdot 10^{-6}$ |
| 0.3 | $8.57 \cdot 10^{-3}$ | $3.10 \cdot 10^{-3}$ | $8.21 \cdot 10^{-4}$ | $4.27 \cdot 10^{-4}$ | $2.18 \cdot 10^{-4}$ | $8.82 \cdot 10^{-5}$ | $4.43 \cdot 10^{-5}$ | $2.22 \cdot 10^{-5}$ | $8.88 \cdot 10^{-6}$ | $4.44 \cdot 10^{-6}$ |
| 0.35 | $1.57 \cdot 10^{-2}$ | $6.69 \cdot 10^{-3}$ | $1.88 \cdot 10^{-3}$ | $9.89 \cdot 10^{-4}$ | $5.07 \cdot 10^{-4}$ | $2.06 \cdot 10^{-4}$ | $1.04 \cdot 10^{-4}$ | $5.19 \cdot 10^{-5}$ | $2.08 \cdot 10^{-5}$ | $1.04 \cdot 10^{-5}$ |
| 0.40 | 2.49 | $1.21 \cdot 10^{-2}$ | 3.63 | $1.92 \cdot 10^{-3}$ | $9.92 \cdot 10^{-4}$ | 4.04 | 2.03 | $1.02 \cdot 10^{-4}$ | $4.09 \cdot 10^{-5}$ | 2.05 |
| 0.5 | 4.89 | 2.87 | 9.63 | $5.22 \cdot 10^{-3}$ | $2.72 \cdot 10^{-3}$ | $1.12 \cdot 10^{-4}$ | $5.63 \cdot 10^{-4}$ | 2.83 | $1.13 \cdot 10^{-4}$ | $5.68 \cdot 10^{-5}$ |
| 0.6 | 7.83 | 5.21 | $1.94 \cdot 10^{-2}$ | $1.08 \cdot 10^{-2}$ | $5.68 \cdot 10^{-3}$ | $2.35 \cdot 10^{-3}$ | $1.19 \cdot 10^{-3}$ | 5.97 | 2.40 | $1.20 \cdot 10^{-4}$ |
| 0.7 | $1.11 \cdot 10^{-1}$ | 8.07 | 3.32 | 1.88 | $1.00 \cdot 10^{-2}$ | 4.18 | 2.12 | $1.07 \cdot 10^{-3}$ | 4.28 | 2.14 |
| 0.8 | 1.46 | $1.13 \cdot 10^{-1}$ | 5.06 | 2.93 | 1.58 | 6.64 | 3.38 | 1.70 | 6.84 | 3.43 |
| 0.9 | 1.83 | 1.47 | 7.13 | 4.22 | 2.31 | $9.77 \cdot 10^{-3}$ | 4.98 | 2.51 | $1.01 \cdot 10^{-3}$ | 5.07 |
| 1.0 | 2.19 | 1.83 | 9.50 | 5.73 | 3.18 | $1.36 \cdot 10^{-2}$ | $6.93 \cdot 10^{-3}$ | 3.50 | 1.41 | 7.07 |
| 1.5 | 3.98 | 3.63 | 2.40 | 1.60 | 9.41 | 4.19 | 2.17 | $1.11 \cdot 10^{-2}$ | 4.48 | $2.25 \cdot 10^{-3}$ |
| 2.0 | 5.60 | 5.29 | 4.01 | 2.89 | 1.80 | $8.38 \cdot 10^{-2}$ | 4.42 | 2.27 | 9.22 | 4.64 |
| 2.5 | 7.02 | 6.75 | 5.55 | 4.26 | 2.81 | $1.36 \cdot 10^{-1}$ | $7.29 \cdot 10^{-2}$ | 3.78 | $1.54 \cdot 10^{-2}$ | 7.77 |
| 3.0 | 8.29 | 8.04 | 6.95 | 5.62 | 3.89 | 1.96 | $1.07 \cdot 10^{-1}$ | 5.58 | 2.29 | $1.16 \cdot 10^{-2}$ |
| 3.5 | $9.42 \cdot 10^{-1}$ | $9.20 \cdot 10^{-1}$ | 8.22 | 6.92 | 5.00 | 2.62 | 1.45 | 7.63 | 3.15 | 1.59 |
| 4.0 | 1.04 | 1.02 | $9.36 \cdot 10^{-1}$ | $8.12 \cdot 10^{-1}$ | 6.09 | 3.32 | 1.86 | 9.89 | 4.11 | 2.08 |
| 5.0 | 1.22 | | 1.13 | 1.03 | $8.20 \cdot 10^{-1}$ | 4.78 | 2.77 | $1.50 \cdot 10^{-1}$ | 6.28 | 3.19 |
| 6 | 1.37 | | 1.30 | 1.21 | 1.01 | 6.28 | 3.74 | 2.06 | 8.73 | 4.45 |
| 7 | 1.51 | | 1.44 | 1.36 | 1.18 | 7.76 | 4.76 | 2.66 | $1.14 \cdot 10^{-1}$ | 5.84 |
| 8 | 1.62 | | 1.56 | 1.50 | 1.34 | $9.19 \cdot 10^{-1}$ | 5.80 | 3.29 | 1.43 | 7.34 |
| 9 | 1.73 | | 1.68 | 1.62 | 1.47 | 1.06 | 6.84 | 3.95 | 1.73 | 8.94 |
| 10 | 1.82 | | 1.78 | 1.72 | 1.59 | 1.19 | $7.88 \cdot 10^{-1}$ | 4.63 | 2.05 | $1.06 \cdot 10^{-1}$ |
| 15 | 2.20 | | | 2.13 | 2.05 | 1.73 | 1.28 | $8.09 \cdot 10^{-1}$ | 3.78 | 2.00 |
| 20 | 2.47 | | | 2.42 | 2.36 | 2.13 | 1.70 | 1.15 | 5.65 | 3.04 |
| 25 | 2.68 | | | | 2.60 | 2.42 | 2.05 | 1.47 | 7.57 | 4.15 |
| 30 | 2.86 | | | | 2.79 | 2.65 | 2.34 | 1.76 | $9.49 \cdot 10^{-1}$ | $5.30 \cdot 10^{-1}$ |
| 50 | 3.35 | | | | 3.31 | 3.24 | 3.09 | 2.65 | 1.67 | 1.00 |
| 80 | 3.82 | | | | | 3.75 | 3.67 | 3.43 | 2.55 | 1.68 |
| 100 | 4.04 | | | | | | 3.93 | 3.76 | 3.01 | 2.09 |
| 300 | 5.13 | | | | | | 5.10 | 5.06 | 4.91 | 4.47 |
| 600 | 5.82 | | | | | | | 5.79 | 5.73 | 5.60 |
| 900 | 6.23 | | | | | | | | 6.17 | 6.10 |

where $a$ is the well response time.

In dimensionless form:

$$h(t) = \frac{FK}{\pi r_w^2} \frac{r_0^2 S}{4T} \frac{Q}{4\pi T} \exp[-(\beta u)^{-1}] \int_u^\infty \frac{\exp(\beta\xi)^{-1}}{\xi^2} W(\xi)d\xi \tag{8}$$

$$= \frac{Q}{4\pi T} W^\star(u,\beta) \tag{9}$$

where

$\beta = \pi r_w^2 4T/FKr_0^2 \quad S$ = response time factor

It can be shown that as $\beta \to 0$, $W^\star(u,\beta) \to W(u)$ and $h \to s$ as expected. Since the integrals of eq.8 are continuous and uniformly convergent, the order of integration can be changed to yield; with a change of variable and integration:

$$W^\star(u,\beta) = \frac{1}{\beta} \exp[-(\beta u)^{-1}] \int_u^\infty \frac{\exp(1/\beta\xi)}{\xi^2} \int_\xi^\infty \frac{\exp(-\lambda)}{\lambda} d\lambda d\xi \tag{10}$$

$$= \int_u^\infty \frac{\exp(-\lambda)}{\lambda} \left[1 - \exp\left\{\frac{-1}{\beta}\left(\frac{1}{u} - \frac{1}{\lambda}\right)\right\}\right] d\lambda \tag{11}$$

Eq. 11 has been evaluated for various values of $\beta$ ranging from 0.1 to 100 and these are shown in Table I. Using the values from Table I a series of type curves can be drawn (Fig.2). As would be expected the effect of observation-well response time is to delay the time at which a value of drawdown occurs and is most important during the early part of an aquifer test. It also delays the time at which the maximum rate of fall of the observation-well water levels occurs, the time lag becoming greater as the response-time factor increases.

## USE OF THE TYPE CURVES WITH OBSERVATION—WELL RESPONSE TIME

### *Type-curve matching*

These new type curves are used to analyse observation-well aquifer-test data in the usual way by plotting the drawdown vs. time on log-log paper, matching the data plot to the most closely followed type curve from Fig.2 and calculating the aquifer parameters and observation-well response time from the match-point values. This approach will be successful if the range of data is sufficient to uniquely determine the drawdown curve. In practice there may be some uncertainty about which value of $\beta$ gives the best type curve match.

The following example will illustrate the method.

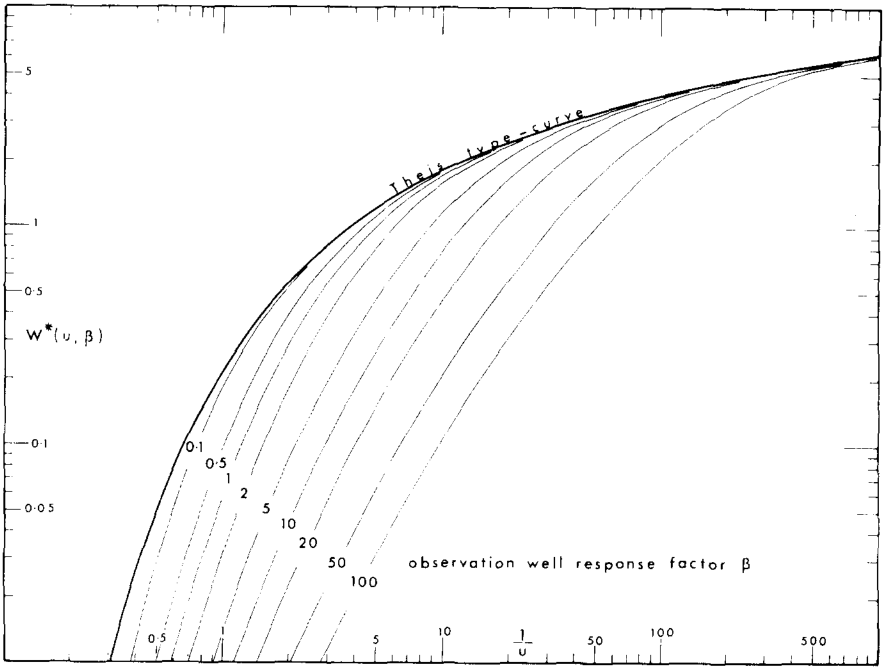


Fig.2. Type curves of the modified well function $W^*(u, \beta)$ for various $\beta$.

An aquifer test was carried out at a site using a pumping well with two observation wells. The pumping well had 50 m of well screen set in 83 m of Late Tertiary unconsolidated sands and sandy marls. The two observation wells were drilled by mud-flush rotary at distances of 80 m (well $A$) and 175 m (well $B$) from the production well. Both had limited wire-wrapped screen (7.3 and 11 m, respectively) set in slightly different parts of the formation. Demudding was not satisfactory but the pumping test was carried out nevertheless.

Fig.3 shows the plot of the drawdown data from well $A$ together with the response-delayed type curve which best fits the data. The value of $\beta$ for this match is 10 which can be converted quickly to a value of response time in minutes by referring to the value along the time axis of the data plot which is overlain by $1/u = 10$ along the type-curve axis. This is equivalent to calculating the values of $T$ and $S$ from the match point and substituting into the definition of $\beta$, i.e.:

$$a = \beta \; (r_0^2 S/4T)$$

The values of $T$ and $S$ are calculated as usual from the chosen match point substituting into the same equations as for a Theis-type curve match. Also marked on Fig.3 are the drawdowns, postulated by the modified analysis, of the aquifer water levels outside the well and the two match points referring to the original type-curve match without observation-well response and the revised type-curve match taking response time into account. The same procedure was carried out for well *B* which had a different response time; the results of analyses are summarised in Table II.

*Slug tests*

The main value of a slug test on the observation well is to verify that the observed deviations of the aquifer test data are not due to some phenomenon other than the observation-well response time. If the well response time for a given observation well from a slug test is in good agreement with that determined from the aquifer test the aquifer parameter results can be treated with greater confidence.

Using the same concepts that underlie the derivation of the well-response type curves and assuming that the volume of water injected for the slug test does not significantly affect the potential in the aquifer around the well, the following expression for the slug test is obtained:

$$H/H_0 = \exp(-t/a) \tag{12}$$

thus when $t = a$:

$$H/H_0 = e^{-1} = 0.37$$

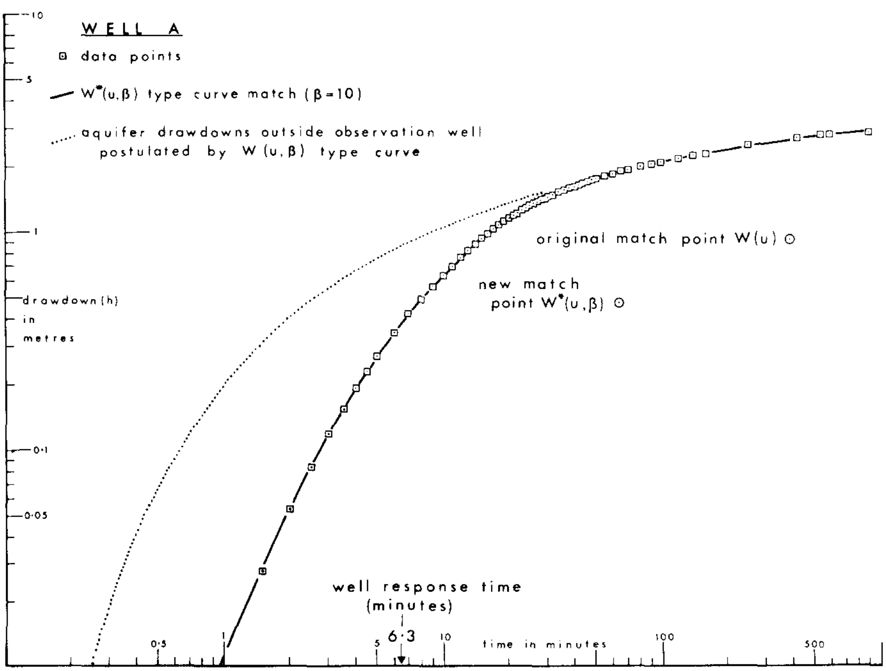


Fig.3. Pumping test results for well *A* with $W^{\star}(u, \beta)$ type curve match.

TABLE II

Aquifer test and slug test results from wells *A* and *B*

| Method of derivation | Parameter | Well *A* | Well *B* |
|---|---|---|---|
| Original type-curve match | transmissivity, $T$ ($m^2$ $day^{-1}$) | 480 | 770 |
| | storativity, $S$ ($\times$ $10^{-4}$) | 9.1 | 6.1 |
| Type-curve match with well response-time lag | transmissivity, $T$ ($m^2$ $day^{-1}$) | 1,020 | 1,060 |
| | storativity, $S$ ($\times$ $10^{-4}$) | 2.9 | 5.1 |
| | value of $\beta$ | 10 | 0.5 |
| | well response time, $a$ (min) | 6.3 | 2.5 |
| Slug test | well response time, $a$ (min) | 8.7 | 2.9 |
| Regional digital model | transmissivity, $T$ ($m^2$ $day^{-1}$) | 930 | 930 |

Slug tests were performed on the two wells *A* and *B* and the data analysed according to Hvorslev (1951) to yield values of well response time. That is, where $H/H_0$ = 0.37, was taken as the value of well response time.

The results show that the consideration of observation-well response time brings the results from the two wells into much closer agreement not only with each other but also with the regional model value. Confidence in the modified type curve matches is enhanced by the reasonable agreement of the well response times rendered by the aquifer test and the slug tests. It is to be noted that storativity is reduced due to the shift of the type curve both downwards (thereby increasing $T$) and also to the left (thereby reducing $S$). The results of the slug tests of wells *A* and *B* were also analysed according to Cooper et al. (1967) and the results showed both wells to have localised values of hydraulic conductivity, $K$, lower by more than a factor of 10, than the average values for the aquifer as a whole. In the light of the drilling method used this is taken to infer significant mud invasion giving rise to a substantial "skin effect".

## CONCLUSIONS

Observation-well response time can have considerable effects upon results from aquifer test analysis. If the parameters upon which the response-time factor, $\beta$, is based are considered it is clear that to make $\beta$ as small as possible the following should be taken into account:

(1) $\beta$ is proportional to the square of the observation-well radius so the diameter of a well penetrating the zone of water-level fluctuation should be reduced to a minimum. Wells which penetrate the zone of water-level fluctuation with a dug upper part and are in continuity with the aquifer by means of a bored lower part will give the most serious response delays.

(2) $\beta$ is inversely proportional to the square of the distance from the pumping well to the observation well so this distance should be made as large as possible consistent with a reliable result.

(3) Well-response effects are inversely proportional to the aquifer storativity so that they are most unlikely in unconfined-aquifer tests.

(4) The ratio of aquifer transmissivity over "effective well transmissivity" ($T/FK$) is perhaps the most important variable. It is really the only factor which the well designer and drilling contractor have any control over. The designer can vary the ratio of screened length to aquifer thickness (the longer the screen the less the value of $\beta$) and the drilling contractor can alter the local hydraulic conductivity considerably. Poor drilling techniques and inadequate development can lead to observation wells with a large "skin effect", i.e. a reduction in the factor $FK$.

Although a well response-time effect can be foreseen and, in part at least, reduced the actual completion and development of an observation well should be checked by performing a slug test upon it. In other branches of experimental science the calibration of measuring devices is considered essential. It is therefore suggested that for all confined-aquifer tests the slug testing of observation wells should become standard practice.

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NOTATION

List of symbols

| | | | |
|---|---|---|---|
| $a$ | = | $\pi r_w^2/FK$ = well response time ("hydrostatic time lag") | T |
| $\beta$ | = | $\pi r_w^2/FK \cdot 4T/r_0^2 S$ = well response time factor | dimensionless |
| $T$ | = | aquifer transmissivity | $L^2\ T^{-1}$ |
| $S$ | = | aquifer storativity | dimensionless |
| $s$ | = | aquifer drawdown | L |
| $t$ | = | time since start of test | T |
| $r_0$ | = | distance from pumping well to observation well | L |
| $Q$ | = | pumping rate from pumping well | $L^3\ T^{-1}$ |
| $h$ | = | observation well drawdown | L |
| $H$ | = | difference between well and aquifer water or piezometric levels | L |
| $Q'$ | = | flow rate from observation well during slug test | $L^3\ T^{-1}$ |
| $F$ | = | a shape factor for an observation well | L |
| $K$ | = | local value of hydraulic conductivity for an observation well | $L\ T^{-1}$ |
| $r_w$ | = | radius of observation well | L |
| $\lambda$, $\eta$ and $\xi$ | | variables of integration | |
| $W(u)$ | = | Theis well function | |
| $W^*(u,\beta)$ | = | response time modified well function | |

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